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Homework 2

1.

The answer is A. The Pearson correlation range is between -1 and 1.

2.

I would suggest that the researchers also perform a Spearman correlation test because it performs a little better than the Pearson correlation when working with a small sample size. I would also suggest constructing a linear model, which might better illuminate why there is a difference in significance between the male group and female group. Though the lectures do not go into detail of interaction terms, an interaction between Sex and Brain Size might be a good idea.

3.

The reasoning is understandable. Log-rank tests are most powerful when the hazard ratio is constant over time. This appears to be the case for the first plot, which appears to have the delayed effect. The second plot appears to have diminished effect, and in such cases weighted Log-rank test is effective. *However*, it is not appropriate to select a test after analyzing the survival curves. It would have been more appropriate to decide prior to seeing the survival curves, therefore I do not quite agree with my collaborator in this instance.

4.

The answer is B.

5.

The statistical considerations in a clinical trial are primary objectives, primary outcome of interest, target study population, and the study design. Other considerations are sample size and power, interim analysis, stopping rules, data management, and plan for analysis. The primary objective involves a key question that one should build the study around and also to determine sample size. The primary outcome is the variable that we measure during the clinical trial, perhaps blood pressure,

height, asthma severity, etc. The target population is important to specify to frame the objective around. If we are interested in understanding the effect of age on cardiovascular disease in the elderly, it is important to consider when contextualizing the trial results. And of course, study design, which is a very important consideration and will have a very great effect on the results and interpretation of the results. The different approaches (rule based, model based, model assisted) ought to be used with intention, and selecting a design that fits the objective and nature of the study is extremely important.

6.

The answer is B.